
6-30-2026

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Recommended Citation

Solikhah, Badingatus (2026) "DOES SUSTAINABILITY REPORT ASSURANCE MATTER? SYMBOLIC VERSUS SUBSTANTIVE ESG GREENWASHING," *Jurnal Akuntansi dan Keuangan Indonesia*: Vol. 23: No. 1, Article 4.

DOI: 10.7454/jaki.v23i1.2206

Available at: <https://scholarhub.ui.ac.id/jaki/vol23/iss1/4>

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DOES SUSTAINABILITY REPORT ASSURANCE MATTER? SYMBOLIC VERSUS SUBSTANTIVE ESG GREENWASHING

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ARTICLE INFO

Keywords:

Greenwashing
ESG Disclosure
Sustainability Report Assurance
Institutional Pressure
Corporate Governance

Received: 31 August 2025

Received in revised form:

1. 25 November 2025

2. 17 January 2026

3. 4 March 2026

Accepted: 10 March 2026

Availabe online: 1 April 2026

ABSTRACT

Background: Greenwashing reduces the credibility of ESG disclosure by creating a gap between reported information and actual performance. Taiwan offers a relevant context because ESG disclosure is partially mandatory, while sustainability assurance is voluntary. This study investigates how sustainability assurance and institutional pressure influence greenwashing and whether governance quality strengthens these relationships. **Method:** This study uses panel data of non-financial firms listed on the TWSE and OTC markets from 2017 to 2023. Greenwashing is measured using standardized ESG ratings and ESG controversies, and a peer-relative measure based on Bloomberg and TEJ database is added for robustness. **Findings:** The results show that sustainability assurance and institutional ownership are negatively associated with greenwashing, suggesting that external verification and investor monitoring enhance the credibility of ESG reporting. Governance quality reinforces these effects, indicating that stronger governance structures respond better to external pressure and are less likely to engage in symbolic ESG practices. These results provide practical insight for regulators and investors in markets with partial or voluntary ESG requirements by showing which mechanisms can help reduce misleading ESG claims. **Conclusion:** Sustainability assurance, investor oversight, and governance quality jointly reduce opportunistic ESG practices. **Novelty/Originality of this article:** This study introduces a refined greenwashing measure using ESG ratings and controversies and provides rare evidence on how assurance and governance interact to reduce greenwashing in a partially mandatory disclosure context.

<https://doi.org/10.7454/jaki.v23i1.2206>

INTRODUCTION

The importance of sustainability has grown rapidly in recent years, supported by the expansion of consumer and capital markets for green products, services, and firms (Delmas and Burbano 2011). Many companies now integrate green strategies into their communication to present themselves as responsible and sustainable organizations (Galletta et al. 2024). National and international commitments to reducing environmental impacts have also created strong incentives for firms to demonstrate responsibility in their operations (Eccles and Krzus 2010; Ghitti et al. 2023). In this context, environmental, social, and governance (ESG) performance

has become an essential signal of long-term orientation, influencing both financial markets and public trust. ESG disclosure is therefore viewed as a key mechanism to communicate commitments and achievements to stakeholders, and at the same time to reduce information asymmetry between firms and the public. Environmental performance in particular has emerged as a central element of modern business legitimacy (Huang et al. 2020; Yue and Li 2023). Stakeholder activism further reinforces this trend, placing pressure on companies to adopt credible ESG practices and maintain transparency (Lee and Raschke 2023).

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As investors, regulators, and society become more sensitive to sustainability issues, firms are increasingly expected to provide comprehensive ESG disclosures that enhance accountability, strengthen relationships with stakeholders, and support long-term competitiveness (Torelli et al. 2020).

Despite its growing importance, ESG disclosure does not always provide an accurate picture of corporate practices. Prior research shows that the signals communicated to stakeholders may differ substantially from actual performance, raising doubts about the reliability of such disclosures (Du 2015; Lee and Raschke 2023). The heavy reliance on self-reported ESG data further increases this risk, as companies can selectively emphasize commitments while minimizing real shortcomings (Galletta et al. 2024; Testa et al. 2018). This situation has contributed to the spread of ESG greenwashing, where firms exaggerate or distort their environmental achievements to secure reputational or financial benefits (Delmas and Burbano 2011; Marquis et al. 2016). Many firms have increased the volume of their sustainability communication, focusing heavily on projecting a green image rather than demonstrating measurable outcomes. The absence of independent assurance, behavioral challenges within companies, and the lack of unified regulatory guidelines further complicate efforts to ensure disclosure accuracy (Friede 2019; Tatomir et al. 2023; Yu et al. 2020). Verifying ESG disclosure and scores also presents difficulties, as methodologies often differ across rating agencies, leading to debates on whether the scores truly represent corporate responsibility (Li et al. 2023). While investors, regulators, and society continue to increase their demands for sustainability reporting, these obstacles create opportunities for firms to engage in selective, incomplete, or misleading disclosures. As a result, the rapid growth of ESG disclosure has been accompanied by

persistent concerns about credibility, reinforcing the need for closer examination of the drivers of greenwashing behavior.

There are several factors that may motivate and alter a company's decision to engage in greenwashing practices (Delmas and Burbano 2011; Sensharma et al. 2022; Yu et al. 2020). Sustainability assurance aims to strengthen ESG disclosure credibility, yet prior findings on its ability to curb opportunism and reputational risks are inconclusive. García-Sánchez et al. (2025) find that firms often adopt assurance in response to ESG controversies. Yet its role may be more symbolic than substantive, as outcomes depend on contextual factors such as firm visibility, controversy severity, and information asymmetry (Birkey et al. 2016; Fernandez-Feijoo et al. 2016). This raises questions about whether assurance can truly limit the risk of ESG greenwashing. At the same time, research on the drivers of greenwashing highlights the importance of external pressures. Evidence shows that shareholder activism may even encourage such practices when firms attempt to meet stakeholder expectations superficially (Kim and Lyon 2015). The specific influence of institutional investors on greenwashing, however, remains underexplored, even though they increasingly rely on ESG information to assess risks and opportunities (Yu et al. 2020). Understanding whether institutional ownership constrains or facilitates greenwashing therefore offers a valuable extension to existing studies (Kim and Lyon 2015; Sensharma et al. 2022; Yu et al. 2020). Prior research also suggests that corporate governance shapes the likelihood of greenwashing (Sensharma et al. 2022; Yu et al. 2020). Yet, while governance quality is often treated as a direct determinant of ESG outcomes, little is known about its moderating role in strengthening or weakening the effects of assurance and institutional pressure.

Building on these gaps, this study aims to examine how sustainability

assurance and institutional ownership influence ESG greenwashing and whether corporate governance quality moderates these relationships. The analysis is conducted in the Taiwanese context, where ESG disclosure has become partially mandatory. This setting provides a unique opportunity to explore how external mechanisms and governance interact in an emerging market with evolving regulatory requirements, offering insights that are relevant for both policymakers and international investors. Accordingly, this study addresses the following research questions: (1) Does sustainability report assurance and institutional ownership reduce ESG greenwashing? (2) Does corporate governance quality moderate the effect of sustainability report assurance and institutional ownership on ESG greenwashing?

This study contributes to the literature in several important ways. First, it introduces a refined measure of greenwashing by comparing ESG disclosure ratings with ESG controversies, thereby capturing the gap between symbolic claims and actual performance. In line with recent work by Cao et al. (2022) and Xia et al. (2023), which employ environmental penalties as an objective benchmark, this approach provides a more reliable assessment than the traditional reliance on self-reported ESG data. Second, while sustainability assurance has rarely been examined in prior greenwashing research, the findings demonstrate that it plays a significant role in mitigating opportunistic reporting. Third, it expands prior work on corporate governance by examining governance quality as a moderating variable—an angle that has rarely been tested—thereby offering new evidence on how governance mechanisms interact with external monitoring forces in shaping ESG disclosure practices. Collectively, these contributions enrich the greenwashing literature by providing new insights into measurement, external assurance, and governance dynamics within

the context of an emerging market.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature and develops the hypotheses. Section 3 describes the research design, data, and variable measurement. Section 4 presents empirical results, including robustness and endogeneity tests. Section 5 discusses the findings in light of prior studies and theoretical perspectives. Finally, Section 6 concludes the practical implications and outlines its limitations and directions for future research.

LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Legitimacy Theory

Legitimacy theory explains that firms seek social approval to sustain their operations within the institutional environment. Organizations must align their activities and disclosures with societal expectations to maintain legitimacy and secure continued stakeholder support (Delmas and Burbano 2011). In the context of ESG reporting, sustainability disclosure is an important instrument used by companies to communicate conformity with environmental and social norms (Delmas and Burbano 2011; Lyon and Montgomery 2015). When stakeholders increasingly demand responsible behavior, companies may respond either substantively or symbolically, by emphasizing favorable disclosure without meaningful changes (Lee and Raschke 2023). Prior studies on greenwashing highlight that firms sometimes use sustainability reporting strategically to preserve legitimacy, especially when actual performance does not fully meet external expectations (Bernini and La Rosa 2023). Moreover, CSR decoupling context suggests that legitimacy can be maintained through formal mechanisms even when substantive alignment is limited (Gull et al. 2023b). Based on this perspective,

legitimacy theory provides a useful foundation for understanding why firms engage in ESG disclosure and under what conditions such disclosure may reduce or instead facilitate greenwashing behavior.

Understanding Greenwashing: ESG Disclosure and Legitimacy

Greenwashing is closely linked to the broader development of ESG disclosure, which has grown in importance since investors began considering environmental and social practices in the 1970s (Richardson 2009). Global initiatives such as the UN Principles for Responsible Investment (PRI 2011) and the Global Reporting Initiative (GRI) have further encouraged companies to disclose their ESG practices, making sustainability reporting a central element of corporate communication (Lokuwaduge and De Silva 2022; Yue and Li 2023). ESG disclosure has been shown to improve investor relations and corporate value (Dhaliwal et al. 2011; Xia et al. 2023). However, the lack of standardized frameworks and the difficulties in verifying ESG claims create space for misleading practices (Friede 2019; Khan et al. 2021; Tatomir et al. 2023). In this context, some firms turn to greenwashing, strategically exaggerating or misrepresenting their sustainability performance to gain legitimacy and stakeholder approval (Delmas and Burbano 2011). Greenwashing may be seen as advantageous for companies but ultimately harmful to society, as CSR decoupling practices are value-destroying and damage corporate legitimacy (Gull et al. 2023a; Testa et al. 2018). According to legitimacy theory, such practices often fail to achieve lasting legitimacy because they do not substantively comply with institutional expectations and regulatory requirements (Delmas and Burbano 2011; Li et al. 2022; Lyon and Montgomery 2015).

The Role of Sustainability Report Assurance in Constraining Greenwashing

Sustainability assurance is designed to function as an external monitoring mechanism that improves the credibility of CSR or ESG disclosures and reduces information asymmetry between firms and stakeholders (Ballou et al. 2018; Healy and Palepu 2001). When a firm decides to assure its sustainability reports, it seeks to send a reliable signal to the capital market and strengthen stakeholder confidence in its ESG commitments (Birkey et al. 2016; García-Sánchez et al. 2025). Assurance is expected to provide independent oversight by verifying the reliability of disclosed information and ensuring that sustainability reports offer a “complete and accurate” picture of a firm’s ESG performance (Ballou et al. 2018; Pflugrath et al. 2011). Experimental evidence from Australia, the United Kingdom, and the United States further demonstrates that assurance of CSR information significantly increases its perceived credibility, particularly among financial analysts (Pflugrath et al. 2011). These findings indicate that assurance can reduce stakeholder skepticism and enhance the trustworthiness of ESG reporting. From the perspective of legitimacy theory, assurance represents a strategy through which firms attempt to maintain or restore social acceptance by aligning their disclosures with stakeholder expectations. However, if assurance is used only symbolically, it may fail to generate substantive legitimacy, as it addresses appearances rather than genuine ESG performance (Delmas and Burbano 2011; Lyon and Montgomery 2015).

Despite these benefits, prior research also highlights the potential symbolic use of assurance. Some firms may adopt assurance mainly as a reputational buffer, prioritizing compliance with external expectations

rather than pursuing substantive ESG improvements (Birkey et al. 2016; García-Sánchez et al. 2022). This paradox suggests that assurance can both legitimize misleading disclosures and at the same time provide credibility, depending on how it is implemented (Fernandez-Feijoo et al. 2016; García-Sánchez et al. 2021). In Taiwan, where sustainability report assurance remains voluntary, obtaining external verification plays an important role in reducing doubts about the accuracy of ESG information and addressing the risk of greenwashing. Overall, assurance serves as an important mechanism to improve transparency and accountability in ESG disclosure (García-Sánchez et al. 2025; Pflugrath et al. 2011). Based on this reasoning, we expect that firms engaging external assurance will provide more credible and higher-quality ESG disclosures, thereby reducing the likelihood of greenwashing behavior. Therefore, hypothesis 1 is proposed as follows:

H1: Firms obtaining sustainability report assurance are negatively associated with greenwashing activities

Institutional Pressure and Its Impact on Greenwashing

According to Kim and Lyon (2015), shareholder pressures are a key factor in determining whether to use a greenwashing approach. Yu et al. (2020) and Sensharma et al. (2022) specifically explore the potential of institutional ownership to mitigate greenwashing practices. Yu et al. (2020) investigate 47 countries over the period 2012 to 2016. They found that a higher percentage of institutional ownership is associated with reduced greenwashing activities. Their results suggest that institutional investors play a crucial role in limiting unethical corporate behavior by providing stronger oversight. They also emphasize that comprehensive ESG data disclosed in sustainability reports can reduce information asymmetry among stakeholders, thereby constraining managers from engaging in greenwashing

through greater monitoring and scrutiny. In the Indian context, Sensharma et al. (2022) also identified a negative relationship between institutional ownership and greenwashing. Their findings show that financial institutions influence greenwashing scores by channeling capital into sustainable projects such as electric vehicles and solar energy. These results highlight the importance of ESG-based investment decisions in shaping corporate behavior and promoting accountability.

Similarly, Testa et al. (2018) argue that capital investors may encourage firms to adopt environmental practices when shareholder environmental awareness is high, reinforcing the role of institutional investors in supporting genuine ESG commitment. Additionally, Delmas and Burbano (2011) caution that “brown firms,” facing strong investor and consumer expectations to appear environmentally friendly, may have incentives to overstate their performance and engage in greenwashing when regulatory consequences are weak. Taken together, these studies suggest that institutional investors can be a powerful force in curbing greenwashing, but their influence may vary depending on firm characteristics and external regulatory environments. Thus, this study proposes the second hypothesis as follows:

H2: Firms with higher institutional ownership are negatively associated with greenwashing activities.

The Moderating Role of Corporate Governance Quality

Corporate governance mechanisms are widely recognized as critical determinants of disclosure quality and corporate accountability. Prior studies demonstrate that firm-level governance elements—such as independent directors, board composition, and institutional ownership—play a significant role in influencing greenwashing practices (Sensharma et al. 2022; Yu et al. 2020). However, much of the existing research has

focused on single attributes of governance, such as board gender diversity (Eliwa et al. 2023; Huang et al. 2022; Zahid et al. 2023) or CEO power (Shahab et al. 2022), leaving limited understanding of the broader impact of comprehensive governance quality. Gull et al. (2023a), for instance, show that the effect of board gender diversity on CSR decoupling varies depending on governance quality, underscoring that stronger governance frameworks generally support greater transparency. Building on this line of reasoning, this study employs an overall measure of governance quality to capture both internal and external mechanisms, extending prior research that often relies on partial indicators. From the perspective of legitimacy theory, higher governance quality should strengthen the credibility of external monitoring mechanisms such as sustainability report assurance and institutional ownership, ensuring that these pressures result in substantive rather than symbolic ESG disclosures.

Moreover, research also emphasizes that corporate governance can serve both enabling and constraining roles in disclosure strategies. On one hand, boards of directors and external stakeholders can encourage disclosure practices that align with societal expectations, while on the other hand governance structures also function as control instruments to prevent opportunistic behavior (Bernini and La Rosa 2023; Velte 2023). Professional accountants and auditors likewise stress that embedding ESG criteria in governance and risk management systems is an effective way to constrain greenwashing (KPMG 2020). Thus, governance quality can amplify the effect of sustainability report assurance by providing robust internal oversight that complements external verification, leading to more credible ESG disclosures. Similarly, institutional pressure is likely to be more effective in curbing greenwashing when reinforced by strong governance systems, as boards and committees translate external

monitoring into substantive corporate practices. Taken together, these arguments suggest that corporate governance quality enhances the effectiveness of both assurance and institutional ownership in reducing greenwashing.

H_{3a}: The negative link between sustainability report assurance and greenwashing activities is more pronounced in firm with strong governance quality.

H_{3b}: The negative link between institutional pressure and greenwashing activities is more pronounced in firm with strong governance quality.

RESEARCH METHODOLOGY

Data and Sample Selection

This study is conducted in the Taiwanese context, which provides a unique setting for examining corporate greenwashing. Taiwan has progressively strengthened its ESG disclosure requirements and introduced partially mandatory reporting for specific categories of firms. The sample consists of non-financial companies listed on the Taiwan Stock Exchange (TSE) and the Over-the-Counter (OTC) markets, covering the period 2017 to 2023. Data were obtained from the Taiwan Economic Journal (TEJ), a widely used and reliable source for financial and non-financial information. The year 2017 is chosen as the starting point because the Taiwan Stock Exchange (TWSE) released the Global Reporting Initiative (GRI) Standards in Traditional Chinese, which encouraged broader adoption and greater consistency in ESG reporting practices among Taiwanese companies.

The initial sample comprised 15,984 firm-year observations, further filtered based on data availability and completeness, as presented in Table 1. First, 958 observations were excluded due to missing ESG data, and a further 1,123 were

Table 1
Sample Selection

Description	Number of Observations
Initial sample for non-financial observations	15,984
Excluded: Lack of ESG data	958
Excluded: Insufficient data for other variables	<u>1,123</u>
Final firm-year observations	13,903

removed because lacked information to measure other variables included in the model. The final dataset comprises 13,903 firm-year observations, representing 2,170 unique companies over the 7-year study period.

Variables Definitions

Dependent Variable

The dependent variable in this study is greenwashing, defined as the gap between a firm's actual ESG performance and its reported claims of environmental responsibility. Greenwashing occurs when companies present themselves as environmentally responsible while, in reality, failing to take substantive actions to reduce their negative environmental impact or to promote social welfare. In this study, greenwashing is measured by comparing standardized ESG ratings with standardized ESG controversies, both obtained from the TEJ database. The raw ESG rating ranges from 1 to 7 (equivalent to A+ to C-) and is calculated from 16 indicators covering environmental, social, and governance aspects. TEJ uses a weighted scoring method that is adjusted for each industry, so the ESG ratings can be compared fairly across different sectors. Meanwhile, the ESG controversies score ranges from 0 to 100, and records negative events such as violations or penalties, where a lower score indicates more serious ESG issues. A large gap, reflected in a high ESG rating combined with a low ESG controversies score, indicates potential greenwashing. This situation suggests that a company reports strong ESG performance while still

facing significant ESG-related problems. The variable is then operationalized using the following formula:

$$GW_{i,t} = \left[\frac{(ESGRating_{i,t} - \overline{ESGRating})}{\sigma ESGRating} \right] - \left[\frac{(ESGCont_{i,t} - \overline{ESGCont})}{\sigma ESGCont} \right] \dots\dots\dots (1)$$

Independent Variables

The first independent variable is sustainability assurance. This variable is measured using a dummy indicator that takes the value of 1 if a company's sustainability report has obtained external assurance or verification from an independent third party, and 0 otherwise. External assurance is considered an important governance mechanism as it provides additional credibility to ESG disclosures and reduces information asymmetry between firms and stakeholders (García-Sánchez et al. 2025; Pflugrath et al. 2011). By relying on third-party validation, assurance enhances the reliability of reported sustainability information and is expected to constrain managerial discretion in greenwashing practices.

The second independent variable is institutional pressure, which is proxied by institutional ownership. It is calculated as the ratio of shares held by institutional investors to the total number of outstanding shares. Institutional investors are recognized as influential stakeholders who can exert monitoring power and discipline over corporate management, given their significant financial stakes and demand for transparent and credible ESG practices (Sensharma et al. 2022; Yu et al. 2020). A

higher proportion of institutional ownership reflects greater external pressure on firms to align with responsible investment standards, thereby reducing the likelihood of opportunistic behaviors such as greenwashing.

Moderator Variable

This study employs governance quality, which captures both internal and external governance mechanisms, as the moderator variable. The measure is based on the Taiwan Corporate Governance Evaluation conducted annually by the Securities and Futures Institute (SFI). The SFI evaluation is mandated by the Taiwan Stock Exchange and Taipei Exchange, and it assesses listed companies using a comprehensive set of indicators aligned with the OECD Principles of Corporate Governance. These indicators cover multiple dimensions, including board structure and independence, shareholder rights, disclosure transparency, and corporate social responsibility practices. For this study, we use the composite corporate governance (CG) score provided in the SFI evaluation as a proxy for governance quality. The CG score ranges from 0 to 100, where a higher score reflects stronger governance structures and greater commitment to transparency and accountability. The score is grounded in OECD principles and incorporates both domestic and international governance standards, global developments, and relevant regulatory guidelines. The SFI evaluation uses a binary “yes” or “no” approach based on publicly available information, ensuring transparency and objectivity in the scoring process. Governance quality is expected to shape how firms respond to external monitoring mechanisms such as sustainability assurance and institutional pressure.

Econometric Model

This study constructs an econometric model to test the proposed hypotheses. All variables, except for

dummy variables, are winsorized at the 1st and 99th percentiles to reduce the influence of outliers in the regression analysis. Consistent with García-Sánchez et al. (2022); and García-Sánchez et al. (2021), a fixed-effects linear regression model for panel data is employed to address unobserved heterogeneity across firms and over time. In line with relevant prior studies, several control variables are also incorporated, namely profitability (ROA), cash flow (CF), leverage (DER), firm size (SIZE), and firm age (AGE). We include firm fixed effects and year fixed effects, and cluster standard errors at the firm level to account for both cross-sectional differences and temporal variations.

Regression Model:

$$GW_{i,t} = \beta_0 + \beta_1 Assurance_{i,t} + \beta_2 Ins_{Pres_{i,t}} + \beta_3 CG_{i,t} + \beta_4 Assurance * CG_{i,t} + \beta_5 Ins_Pres_{i,t} * CG_{i,t} + Controls_{i,t} + Firm\ Effect + Year\ Effect + \varepsilon_{i,t} \dots\dots\dots (2)$$

RESULT AND ANALYSIS

Descriptive Statistics

Table 2 presents the descriptive statistics of the variables used in this study. The dependent variable, greenwashing (GW), shows a mean value of 0.000, indicating that on average firms in the sample do not exhibit greenwashing activities. However, the distribution suggests heterogeneity, with values ranging from -1.550 to 2.550 and a standard deviation of 0.861. This spread implies that while many firms report balanced ESG performance relative to controversies, there are cases of substantial overstatement as well as understatement of ESG activities. Regarding the independent variables, sustainability assurance (ASSURANCE) has a mean of 0.177, reflecting that only around 18% of firms obtained external assurance for their sustainability reports.

Table 2
Descriptive Statistics

Variables	Mean	Median	Maximum	Minimum	Std. Dev.	Skewness	Kurtosis
GW	0.000	-0.125	2.550	-1.550	0.861	0.635	3.122
ASSURANCE	0.177	0.000	1.000	0.000	0.382	1.693	3.867
INS_PRES	1.865	0.040	100.000	0.000	5.106	10.829	191.097
CG	52.252	53.670	84.410	0.000	13.306	-1.328	6.770
ROA	0.031	0.039	0.290	-0.470	0.104	-1.422	8.304
CF	0.180	0.197	2.210	-4.710	0.811	-2.771	17.994
DER	0.468	0.273	3.630	0.000	0.622	2.644	11.867
SIZE	0.152	0.150	0.224	0.098	0.016	0.635	3.901
AGE	13.826	13.000	61.000	0.000	10.695	1.085	5.086

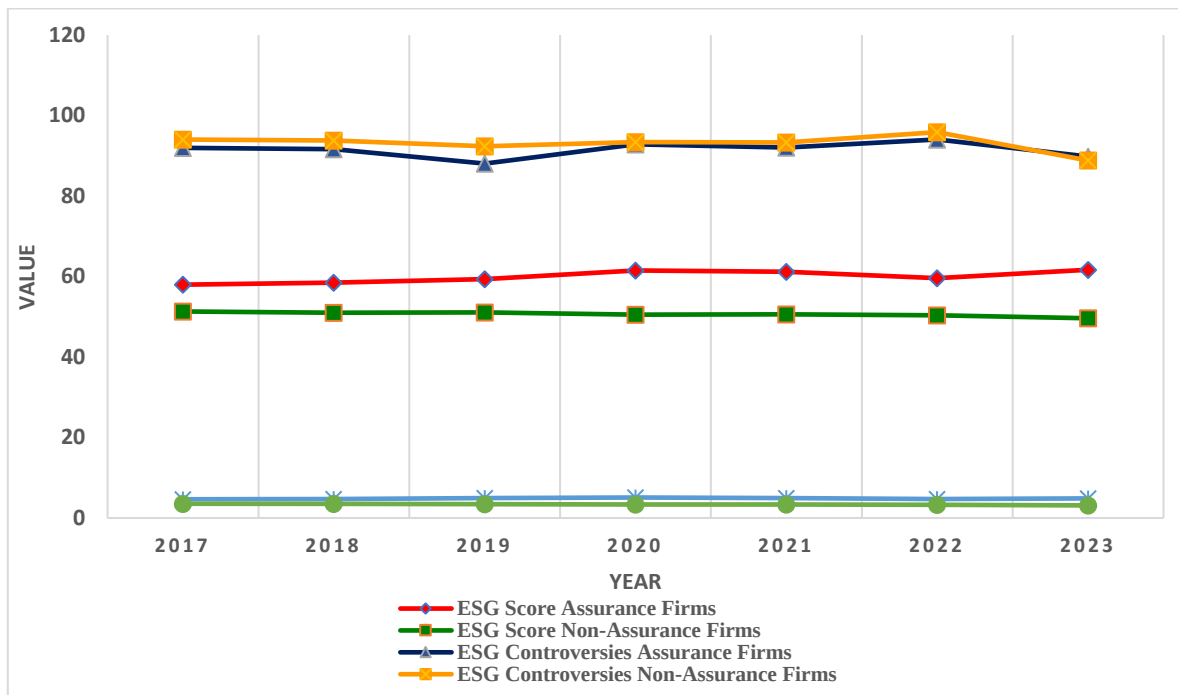


Figure 1
ESG Rating, Score and Controversies between Assurance vs Non-Assurance Groups

Institutional pressure (INS_PRES) shows a mean of 1.865 but with high dispersion, driven by a few firms with very large institutional ownership. The moderator variable, governance quality (CG), averages 52.25, close to the median of 53.67, suggesting moderate governance performance across firms. Control variables also reveal substantial variation: profitability (ROA) is low on average (0.031), leverage (DER) averages 0.468, while firm age (AGE) ranges widely from newly listed firms to those over six decades old.

Figure 1 presents the average ESG Rating, ESG Score, and ESG Controversies for firms with and without sustainability assurance from 2017 to 2023. On average, companies that obtain external assurance consistently report higher both ESG rating and ESG scores compared to their non-assured counterparts. This pattern suggests that assurance is associated with improved disclosure quality and stronger ESG performance as perceived by rating agencies. However, the opposite trend occurs when observing ESG controversies. Firms with assurance exhibit lower controversy scores, which indicates that

Table 3
Correlation Matrix

Variables	GW	ASSUR	INS_PRES	CG	ROA	CF	DER	SIZE	AGE
GW	1.000								

ASSURANCE	0.398	1.000							
	0.000	-----							
INS_PRES	0.194	0.146	1.000						
	0.000	0.000	-----						
CG	0.332	0.169	-0.019	1.000					
	0.000	0.000	0.027	-----					
ROA	0.120	0.054	0.084	0.110	1.000				
	0.000	0.000	0.000	0.000	-----				
CF	0.096	0.062	0.056	0.127	0.567	1.000			
	0.000	0.000	0.000	0.000	0.000	-----			
				-	-	-			
DER	0.056	0.078	-0.013	0.056	0.166	0.077	1.000		
	0.000	0.000	0.135	0.000	0.000	0.000	-----		
SIZE	0.456	0.563	0.323	0.238	0.270	0.191	0.202	1.000	
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-----	
AGE	0.183	0.304	0.083	0.199	0.026	0.083	0.123	0.400	1.000
	0.000	0.000	0.000	0.000	0.002	0.000	0.000	0.000	-----

they are linked to more negative ESG events, violations, or penalties than non-assured firms. This situation shows that there is a gap between what the companies present in their sustainability reports and the real ESG problems they still experience. In other words, although the reports create a positive image, these firms continue to face various ESG challenges in practice.

By showing all three ESG measures together, Figure 1 gives a clearer picture of the differences between assured and non-assured firms. It shows that firms with assurance may look better in their ESG Rating and ESG Score, but at the same time face more serious ESG controversies. This pattern provides early descriptive evidence that is related to the greenwashing issue examined in this study.

To explain this paradox, the literature on CSR assurance provides two contrasting perspectives that help explain these results. On the one hand, assurance is intended to enhance reporting credibility and reduce information asymmetry (Birkey et al. 2016). On the other hand, several studies highlight the symbolic use of assurance, where firms employ third-party

verification mainly as an impression management tool to project transparency and legitimacy rather than to improve substantive performance (Fernandez-Feijoo et al. 2016). Because CSR assurance remains voluntary and is often unregulated, companies may strategically use it as a “credibility cover” (Ballou et al. 2018), thereby masking poor ESG practices behind higher reported scores. This aligns with concerns about “managerial capture” and the lack of standardized assurance procedures, which can compromise the independence and effectiveness of assurance providers (Pflugrath et al. 2011). In this context, sustainability assurance may be used to strengthen the appearance of ESG disclosures, even when the firm still faces significant ESG controversies. This pattern aligns with early signs of greenwashing behavior, which is widely discussed in the literature (García-Sánchez et al. 2021; Zharfpeykan 2021).

Table 3 reports the correlation matrix of the variables used in the study. The first row of each cell presents the correlation coefficients, while the second row provides the corresponding probability

Table 4
Main Regression Results

Variables	Model 1		Model 2	
	Coefficient	t-Statistic	Coefficient	t-Statistic
C	0.029	0.191	1.129	1.367
ASSURANCE	-1.314***	-7.041	-1.299***	-6.661
INS_PRES	-0.021***	-8.088	-0.021***	-8.428
CG	-0.001	-0.410	-0.001	-0.241
ASSURANCE x CG	-0.028***	-9.506	-0.027***	-9.192
INS_PRES x CG	-0.000***	-6.796	-0.000***	-6.668
ROA	-	-	0.252	1.572
CF	-	-	-0.011	-1.172
DER	-	-	0.050	1.826
SIZE	-	-	-7.248	-1.216
AGE	-	-	-0.024	-0.443
Firm Effect	Included		Included	
Year Effect	Included		Included	
N	13,903		13,903	
Adjusted R ²	64.12%		64.18%	
F Statistic	12.394***		12.399***	

Note: ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Significance levels are calculated using White's (1980) heteroskedasticity-adjusted standard errors. Definitions for all variables can be found in the variables definitions sections.

values. Overall, the results indicate no serious correlation among the independent variables, suggesting that multicollinearity is not a concern in the subsequent regression analysis.

Empirical Results

Table 4 presents the main regression results. Model 1 reports the regression without control variables, while Model 2 includes the set of control variables (ROA, CF, DER, SIZE, and AGE). Sustainability assurance is found to have a significant negative association with greenwashing, with coefficients of -1.314 (t-stat = -7.041) in Model 1 and -1.299 (t-stat = -6.661) in Model 2. Similarly, institutional pressure also shows a consistent negative effect on greenwashing, with coefficients of -0.021 (t-stat = -8.088) and -0.021 (t-stat = -8.428), respectively. These results provide strong support for H1 and H2, suggesting that both assurance and institutional ownership play a meaningful role in reducing the likelihood of greenwashing. In contrast, governance quality alone does not show a significant effect, as indicated by

coefficients close to zero (t-stat = -0.410 and -0.241). However, the interaction terms between assurance and governance quality, as well as between institutional pressure and governance quality, are both negative and highly significant (ASSURANCE $\times$ CG: -0.028 , t-stat = -9.506 ; -0.027 , t-stat = -9.192 ; INS_PRES $\times$ CG: -0.000 , t-stat = -6.796 ; -0.000 , t-stat = -6.668). These results indicate that strong governance quality strengthens the negative relationship between assurance, institutional ownership, and greenwashing, thereby supporting H_{3a} and H_{3b}.

Robustness Test

Alternative Measurement of Greenwashing

Different proxies have been used in prior studies to measure ESG greenwashing (Gull et al. 2023a; Li et al. 2023). To ensure the robustness of our findings, this study examines whether the results remain consistent when an alternative proxy is applied. The baseline analysis relies on the TEJ database, where greenwashing is

Table 5
Regression Results Using Alternative Measurement of Greenwashing

Variables	Model 1		Model 2	
	Coefficient	t-Statistic	Coefficient	t-Statistic
C	1.664***	7.381	0.426	0.495
ASSURANCE	-0.051**	-2.140	-0.037**	-2.037
INS_PRES	-0.014*	-1.922	-0.014*	-1.916
CG	-0.041***	-10.652	-0.041***	-18.956
ASSURANCE x CG	-0.003	-0.565	-0.003*	-1.904
INS_PRES x CG	-0.000	-0.564	-0.000*	-1.891
ROA	-	-	-0.075	-0.420
CF	-	-	-0.080***	-2.749
DER	-	-	0.015	0.473
SIZE	-	-	4.133	0.795
AGE	-	-	0.207***	2.884
Firm Effect	Included		Included	
Year Effect	Included		Included	
N	2,671		2,671	
Adjusted R ²	81.12%		81.26%	
F Statistic	25.254***		25.222***	

Note: ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Significance levels are calculated using White's (1980) heteroskedasticity-adjusted standard errors. Definitions for all variables can be found in the variables definitions sections.

captured by the standardized ESG rating minus standardized ESG controversies. As a robustness check, we also adopt a peer-relative greenwashing measure following Yu et al. (2020) and Birindelli et al. (2024). In this alternative specification, ESG disclosure scores are obtained from the Bloomberg database, while ESG performance scores are drawn from TEJ, and the index compares the two by standardizing each relative to the sample mean and standard deviation. The alternative greenwashing score is formally defined as:

$$GW2_{i,t} = \left[\frac{(ESGDis_{i,t} - \overline{ESGDis})}{\sigma_{ESGDis}} \right] - \left[\frac{(ESGPer_{i,t} - \overline{ESGPer})}{\sigma_{ESGPer}} \right] \dots \dots \dots (3)$$

Table 5 reports the regression results using an alternative measure of greenwashing. The findings remain broadly consistent with the main analysis, supporting the robustness of the results. Sustainability assurance continues to show a negative and significant effect on greenwashing (-0.051 , $t = -2.140$; -0.037 ,

$t = -2.037$), confirming that external assurance helps constrain opportunistic disclosure. Institutional pressure also retains its negative association with greenwashing, albeit at weaker levels of significance (-0.014 , $t \approx -1.95$). Governance quality itself is strongly negative and highly significant across both models (-0.041 , $t = -10.652$; -0.041 , $t = -18.956$), indicating that stronger governance structures play a central role in reducing greenwashing. The interaction terms show limited evidence of moderating effects: in Model 2, both $ASSURANCE \times CG$ (-0.003 , $t = -1.984$) and $INS_PRES \times CG$ (-0.000 , $t = -1.899$) are marginally significant at the 10% level, suggesting that governance may modestly strengthen the disciplining effect of assurance and institutional pressure. Overall, the robustness check reinforces the main findings.

Industry Control

It is important to account for industry-specific characteristics, since firms operating in different sectors are

Table 6
Regression Results Using Industry Effect

Variables	Model 1		Model 2	
	Coefficient	t-Statistic	Coefficient	t-Statistic
C	-0.001	-0.004	-3.085***	-19.659
ASSURANCE	-1.313***	-7.058	-0.985***	-6.199
INS_PRES	-0.021***	-8.406	-0.008***	-4.290
CG	-0.001	-0.343	-0.012***	-10.335
ASSURANCE x CG	-0.028**	-9.542	-0.024***	-8.931
INS_PRES x CG	-0.000***	-7.004	-0.000**	-2.035
ROA	-	-	0.158	1.425
CF	-	-	0.002	0.172
DER	-	-	0.004	0.223
SIZE	-	-	15.980***	14.984
AGE	-	-	-0.009	-0.788
Industry Effect	Included		Included	
Year Effect	Included		Included	
N	13,903		13,903	
Adjusted R ²	64.22%		64.26%	
F Statistic	12.208***		12.222***	

Note: ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Significance levels are calculated using White's (1980) heteroskedasticity-adjusted standard errors. Definitions for all variables can be found in the variables definitions sections.

subject to distinct pressures and constraints that influence their sustainability strategies and disclosure practices. As noted by Gull et al. (2023b), the quality of sustainability reporting varies substantially across industries, with environmentally sensitive sectors such as mining and oil and gas often providing more extensive disclosures to address stakeholder expectations. To capture these differences, this study incorporates industry effects into the regression models.

Table 6 presents the regression results after controlling for industry effects. The findings remain consistent with the main analysis. Sustainability assurance continues to exhibit a significant negative association with greenwashing (-1.313 , $t = -7.058$ in Model 1; -0.985 , $t = -6.199$ in Model 2), while institutional pressure also retains its negative and highly significant effect (-0.021 , $t = -8.406$; -0.008 , $t = -4.290$). Governance quality alone shows limited significance in Model 1 but becomes strongly negative in Model 2 (-0.012 , $t = -10.335$). More importantly, the

interaction terms reinforce earlier results: $ASSURANCE \times CG$ remains negative and highly significant (-0.028 , $t = -9.542$; -0.024 , $t = -8.931$), and $INS_PRES \times CG$ also shows a negative moderating effect (-0.000 , $t = -7.004$; -0.000 , $t = -2.035$). Overall, these results confirm the main findings.

Endogeneity Test

To address potential endogeneity concerns, this study follows the approach used in prior research by applying a one-period lag to all explanatory and control variables. This method is commonly used to reduce reverse causality because business decisions in ESG practices often influence each other over time (García-Sánchez et al. 2022; Parra-Domínguez et al. 2021). Following García-Sánchez et al. (2025), we use a one-period lag as the main approach to reduce endogeneity. This helps ensure that the independent variables come before the greenwashing behavior in time. In addition, we continue to use firm fixed

Table 7
Endogeneity Test Results Using Lagged Independent Variables

Variables	Model 1		Model 2	
	Coefficient	t-Statistic	Coefficient	t-Statistic
C	-0.364***	-4.694	-1.701*	-2.181
ASSURANCE _{t-1}	-0.235*	-2.295	-0.178*	-2.019
INS_PRES _{t-1}	-0.009*	-2.426	-0.009*	-2.217
CG _{t-1}	0.006	0.113	-0.005	-0.277
ASSURANCE _{t-1} x CG _{t-1}	-0.001*	-2.283	-0.006***	-6.427
INS_PRES _{t-1} x CG _{t-1}	-0.000*	-2.131	-0.000*	-2.194
ROA _{t-1}	-	-	-0.171	-1.888
CF _{t-1}	-	-	0.007	0.887
DER _{t-1}	-	-	-0.030	-1.765
SIZE _{t-1}	-	-	10.401*	2.240
AGE _{t-1}	-	-	-0.094*	-2.378
Industry Effect	Included		Included	
Year Effect	Included		Included	
N	11,021		11,021	
Adjusted R ²	68.01%		75.58%	
F Statistic	14.177***		17.572***	

Note: ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Significance levels are calculated using White's (1980) heteroskedasticity-adjusted standard errors. Definitions for all variables can be found in the variables definitions sections.

effects to control for unobserved characteristics that remain constant across time, which further helps address potential endogeneity (García-Sánchez et al. 2025). Table 7 reports the results of this endogeneity test. Model 1 presents the regression without control variables, while Model 2 includes the control variables (ROA, CF, DER, SIZE, and AGE). The results from both models remain consistent with the main findings, even though the significance levels decrease. This consistency indicates that the relationships identified in this study are stable and not materially affected by endogeneity concerns.

DISCUSSIONS

The Effect of Sustainability Report Assurance on Greenwashing Behaviors

The regression results provide

strong support for H₁, showing that firms engaging external sustainability assurance are less likely to be involved in greenwashing. This finding aligns with the view that assurance enhances the credibility of ESG disclosures and mitigates information asymmetry by signaling reliability to investors and other stakeholders (Birkey et al. 2016). From the perspective of legitimacy theory, assurance functions as a mechanism for firms to demonstrate conformity with societal expectations, aligning their disclosures with stakeholder expectations and maintain legitimacy through credible ESG reporting. In Taiwan, where sustainability assurance remains voluntary and only about 18% of firms obtain such verification, the results suggest that firms choosing to engage assurance do so in a substantive way, using it to strengthen transparency and accountability rather than merely for symbolic purposes. This interpretation

resonates with prior evidence that external assurance, when adopted voluntarily, can play an important role in reducing opportunistic disclosure and fostering stakeholder trust (García-Sánchez et al. 2025).

At the same time, these findings contribute to the ongoing debate about the substantive versus symbolic role of assurance in sustainability reporting. Earlier studies caution that some companies use assurance primarily for impression management, creating a “credibility cover” without addressing underlying ESG performance (Ballou et al. 2018; Fernandez-Feijoo et al. 2016). Legitimacy theory suggests that such symbolic strategies may offer short-term legitimacy benefits but are fragile, as they do not resolve the gap between disclosed commitments and actual practices. However, the negative and significant effect observed in this study suggests that, at least within the Taiwanese context, assurance appears to constrain greenwashing rather than facilitate it. This result complements recent research showing that external verification can reduce the gap between reported ESG commitments and actual performance, thereby countering CSR decoupling (García-Sánchez et al. 2021). Overall, the evidence underscores the relevance of sustainability assurance as a credible monitoring mechanism that enhances the quality of ESG disclosure and limits managerial discretion in symbolic reporting.

The Effect of Institutional Pressure on Greenwashing Behaviors

The results strongly support H2, indicating that higher levels of institutional ownership are associated with reduced greenwashing practices. This suggests that institutional investors act as effective external monitors, demanding credible and transparent ESG disclosures to protect their investments and reduce information asymmetry (Sensharma et al. 2022; Yu et

al. 2020). From the perspective of legitimacy theory, institutional investors serve as critical stakeholders whose expectations push firms to conform to socially accepted norms of environmental and social responsibility. Their financial influence and monitoring capacity limit managerial discretion, discouraging opportunistic behaviors and promoting substantive sustainability practices.

These findings are consistent with prior evidence that institutional investors play an important role in shaping corporate ESG outcomes. For instance, Sensharma et al. (2022) show that financial institutions in India channel funds toward environmentally sustainable projects, thereby lowering corporate incentives to engage in greenwashing. Similarly, Testa et al. (2018) argue that shareholder environmental awareness can drive firms to adopt genuine environmental practices, while Delmas and Burbano (2011) warn that “brown firms” often face pressure to appear green without substantive actions. In line with these insights, the present study finds that institutional pressure in Taiwan generally strengthens ESG accountability and curbs greenwashing behaviors. Given that the average level of institutional ownership in the sample is relatively low but varies widely, the results highlight the disciplining effect of institutional investors and their potential as a transformative force in improving ESG practices in emerging markets. Within the legitimacy theory framework, institutional investors represent powerful stakeholders whose expectations shape firms’ efforts to secure and maintain legitimacy.

The Moderating Role of Corporate Governance Quality

The results provide strong support for H_{3a} and H_{3b}, showing that governance quality significantly strengthens the negative associations between sustainability assurance and greenwashing as well as between institutional pressure and greenwashing. These findings suggest

that corporate governance mechanisms not only function as direct determinants of disclosure quality but also enhance the effectiveness of external monitoring forces. In line with legitimacy theory, higher governance quality ensures that firms respond to institutional and stakeholder demands with substantive rather than symbolic actions, thereby reducing the risk of greenwashing. This aligns with prior work indicating that comprehensive governance frameworks improve transparency and accountability by constraining opportunistic managerial behavior (Sensharma et al. 2022; Yu et al. 2020). Moreover, Gull et al. (2023b) show that the impact of board-level characteristics on CSR outcomes depends on governance quality, emphasizing that firms with stronger governance structures are better positioned to deliver authentic reporting. From a legitimacy theory perspective, strong governance supports legitimacy maintenance by ensuring that firms align their ESG disclosures with societal expectations through substantive practices rather than symbolic compliance, thereby strengthening stakeholder acceptance and reducing greenwashing risks.

These results also resonate with the dual role of governance described in the literature. Governance can serve as an enabler of disclosure strategies aligned with stakeholder expectations, but it can also act as a control instrument to limit information asymmetry and opportunism (Bernini et al. 2023; Velte 2023). The significant interaction effects observed in this study support the control perspective, where robust governance strengthens the credibility of both sustainability assurance and institutional ownership. Professional guidance also highlights that embedding ESG criteria into governance and risk management systems is critical to constraining greenwashing (KPMG 2020). Taken together, these findings indicate that strong governance structures in Taiwanese firms play a pivotal moderating role,

transforming external monitoring pressures into substantive ESG engagement rather than symbolic compliance. This reinforces the idea that governance quality operates as a complementary mechanism that amplifies the effectiveness of assurance and institutional investors in reducing greenwashing.

CONCLUSION

This study investigates the determinants of ESG greenwashing in the Taiwanese context, where ESG disclosure is partially mandatory and sustainability assurance remains voluntary. Using panel data of non-financial firms from 2017 to 2023, the analysis highlights three key findings. First, sustainability assurance is negatively associated with greenwashing, indicating that external verification helps strengthen the credibility of ESG reporting. Second, institutional ownership exerts a disciplining effect, as higher institutional pressure reduces opportunistic disclosure behaviors. Third, governance quality plays a pivotal moderating role, amplifying the effectiveness of both assurance and institutional pressure in constraining greenwashing. Taken together, these results underscore the importance of complementary internal and external monitoring mechanisms in improving the credibility of ESG practices in emerging markets.

A key limitation of this study is that ESG controversies are measured solely using the TEJ database, which may not fully capture the severity and nature of the underlying events due to data availability constraints. Future research could extend this approach by incorporating alternative sources of controversy data or qualitative assessments to provide a more nuanced understanding of greenwashing practices.

Practical Implications

This study offers several practical implications. For companies, obtaining

sustainability assurance can help protect their reputation, especially when facing ESG controversies. For investors, the presence of assurance and the extent of its coverage can be used to identify which firms are providing more reliable ESG information, as opposed to symbolic reporting. The findings also highlight the important role of institutional investors, whose monitoring can push firms toward more credible ESG practices. Similarly, stronger corporate governance structures enhance the effectiveness of both assurance and investor pressure, ensuring that disclosure quality is not only for appearance but reflects substantive performance. Finally, the results are relevant beyond Taiwan, offering guidance for regulators and firms in emerging markets, particularly in Asian countries, where similar institutional settings exist to improve the credibility of ESG reporting.

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